

Examining the effects of governmental networking with environmental turbulence on the geographic searching of business model innovation generations

Haixu Bao, Chunhsien Wang and Ronggen Tao

Abstract

Purpose – This study aims to explore the relationship between geographic search and business model innovation and proposed a contingent framework to focus on how governmental networking and environment turbulence are interdependent moderate the relationship between geographic search and business model innovation.

Design/methodology/approach – A large-scale questionnaire survey was carried out among the firms in three high-tech parks of the Pearl River Delta, with a total of 287 firms as empirical samples. Hypotheses are tested using ordinary least squares analyzes on hierarchical multiple regression to find out how geographic search can drive business model innovation generations.

Findings – The empirical results showed that the more frequent geographic search is, the more favorable it is for firms to generate innovative business models, and firms may be more effective in geographic searching and business model innovation with better governmental networking. However, the above relationship may be weakened if the environment turbulence in emerging markets is further considered. It was argued that firms must take into account both the positive effects of governmental networking and the negative effects of environmental turbulence in conducting a geographic search for external knowledge resources to generate innovative business models. The study results showed how and why governmental networking can be a key catalyst for firms to generate innovative business models.

Research limitations/implications – This study contributes to the business model innovation literature by documenting the large-scale survey evidence that confirms the practicality of geographic search in the business model innovation generations. The findings advance previous studies in the business model innovation by identifying the moderating roles of governmental network and environment turbulence that predict business model innovation behaviors in the emerging market.

Practical implications – The results indicate that the geographic search can be easily operationalized for external resources acquisitions by managers in generating business model innovation. This has applications for external resource acquisitions on the basis of business model innovation in the emerging China market. In addition, to facilitate the business model innovation generations, the focus should be on critical contingency factors; on the one hand, to promote the continued use of external resources, the focus should be on enhancing benefits such as governmental networking.

Originality/value – The findings extend existing theory in three ways as the original value. First, the results show that geographic search is an important driver of business model innovation generations in an emerging market context. Second, this study is the first to take organizational learning and open innovation perspective to examine geographic search as a boundary-spanning search of external resources in business model innovation generations. Third, this study also explores the moderator role of governmental network and environmental turbulence on how to strengthen or impair the geographic search and business model innovation generations.

Keywords Emerging market, Business model innovation, Environment turbulence, Geographic searching, Governmental networking

Paper type Research paper

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Introduction

As the issue of a business model (BM) has attracted the attention of scholars and practitioners, more and more studies have focused on business model innovation (Amit and Zott, 2001; Spieth *et al.*, 2014). So far, there has been no widely accepted consistent and complete definition of business model innovation (Holm *et al.*, 2013; Teece, 2010; Chin *et al.*, 2020). Although, business model innovation is still a very important concept and a means of communication (Zott *et al.*, 2011) or is considered as a tool for firms to create and obtain value (Chesbrough and Rosenbloom, 2002; Baden-Fuller and Morgan, 2010; Teece, 2010; Oliva *et al.*, 2018; Wang and Chin, 2020). In this study, based on the previous studies, business model innovation was conceptualized as a tool to explain the activities of firms to create and capture value for customers in terms of products, services, processes and even profit models (Chesbrough and Rosenbloom, 2002; Baden-Fuller and Morgan, 2010; Casadesus-Masanell and Ricart, 2010; Teece, 2010; Martins *et al.*, 2015; Oliva *et al.*, 2018). Hence, in this study, business model innovation was defined as a profit model in which firms can develop new products, services and processes and create value for customers in operating activities. Most of the previous studies on business model innovation focused on exposing the relationship between business model innovation and firm performance (Zott and Amit, 2007, 2008), but some problems still need to be solved. Especially, the antecedents driving firms to innovate business models are limited (Foss and Saebi, 2016), and now, the studies on driving firms to innovate business model generation is still in an early stage.

It is very important to understand the antecedents of firms' business model innovation generation because antecedents are the sources for firms to generate new business models and improve competitiveness. In recent years, it has been pointed that the antecedents driving firms to innovate business models are mainly divided into internal and external contingent characteristics (Chin *et al.*, 2020; Foss and Saebi, 2016). Internal contingent characteristics are changes in strategies or dynamic capabilities; and external contingent characteristics are competition changes, technology, interested parties' needs, network sites and the introduction of new opportunities, leading firms to innovate business models (Foss and Saebi, 2016). This study focused on external contingent factors, that is, how firms explore opportunities and obtain resources through boundary-spanning searching for business model innovation; to be more specific, how firms obtain external knowledge resources through boundary-spanning geographic search to achieve innovative business models (Chin *et al.*, 2020; Malhotra, 2005; Zheng *et al.*, 2020). Indeed, geographic boundaries searching is helpful for firms to obtain the resources needed for business innovation (Singh, 2008). In addition, even though boundary-spanning searching is helpful for innovating business model generations, the role of government is still unknown in this relationship. The key insight from extant research is that governments' supports to the operating activities of the firms in emerging markets such as the acquisition of finance and external technology are greatly helpful to their innovative activities (Li and Atuahene-Gima, 2002; Del Giudice and Della Peruta, 2016; Della Peruta *et al.*, 2014), especially, governmental networking provides an important channel to acquire public resources and knowledge (Du *et al.*, 2016), to promote firms' innovative activities. In such cases, the importance of governmental networking to business models is highlighted in the rapidly growing studies on business model innovation. However, it is necessary to understand how governmental networking promotes firms to conduct a geographic search for external resources, so as to effectively guide and strengthen business model innovation, but the empirical studies are still very limited.

To address the above difficulties, we examine if external resources and opportunities from geographic searching positively impact the business model innovation generation, and if so, whether or not the relationships differ according to the governmental networking under a dynamic environment. We specifically develop the argument that although a firm's

geographic search creates the opportunities to engage in innovative business model generation, the firm needs appropriate back-up with governmental networking to fully exploit this opportunity. The most important implication of this argument is that governmental networking helps explain why a firm would depend more on informal ties to cultivating business activities. Thus, identifying conditions under which the joint moderating effect of geographic searching and governmental networking on the business model generation is manifested in the dynamic emerging environment. We believe our study is the first to propose an interesting model to extend the geographic search function and the governmental network effects in innovating business model generations.

As one of the world's largest economies, China has provided an important and idiosyncratic setting to test the contingent value of boundary-spanning geographic search, emerging China market has gradually adopted the market economy for economic development and policy. However, compared with most western economies, the market competition rules in the emerging China economy, are still unpredictable and unclear because the formal institutions supporting the market economy such as effective legal infrastructure, are still developing (Hoskisson *et al.*, 2000). Therefore, the governmental network has become popular to fill the institution gap in markets, so that managers can supplement the formal institutional supports with personal networking and connections (for example, relationships with various levels of government agencies). According to Xin and Pearce (1996), in the Chinese market, relationships as the mechanisms to coordinate transactions have become an important lifeline for business activities. Hence, China's emerging economy was used as the object of the empirical study to explore the effects of governmental networking and environmental turbulence on business model innovation of firms in this market.

We believe our findings contribute to the following aspects. First, this study made specific contributions to the literature, especially the effects of geographic search on business model innovation. The effects of geographic search on innovation have been explored in previous studies (Cantwell and Janne, 1999), but the design of this study allowed us to reveal the field of business model innovation, so as to establish a new theoretical model of external contingency-driven business model innovation generations. It made up the gap in the literature, especially the effects of external contingent factors on business model innovation (Foss and Saebi, 2016). Second, our study tried to explore the effects of governmental network and environment turbulence on the above relationships by developing the contingency theory about the effects of boundary-spanning geographic search on business model innovation. Specifically, we argued that governmental network can effectively help firms to search and acquire geographic resources, so as to realize business model innovation generations, but the benefits may be offset by damages caused by environment turbulence. Third, this is the first study examining how geographic search affects business model innovation generations and whether firms' geographic search directly predicts their business model innovation generations.

Theory and hypothesis development

Geographic search is defined as a way for firms to cross-organizational and geographic boundaries to access new external ideas and knowledge for their business innovations (Sidhu *et al.*, 2007; Ahuja and Katila, 2004). In this study, the geographic search is not only regarded as a searching behavior but also an approach for firms to acquire new technology and knowledge, expand new markets and develop new users and target consumers (Cooper and Inoue, 1996). Geographic searching has become one of the best ways for firms to obtain external knowledge resources. According to Kafouros *et al.* (2018) and Papa *et al.* (2018), the external technical knowledge acquired by firms from many regions not only improves their performance but also enhances their understanding of technology and their abilities to develop new international markets. For instance, Sidhu *et al.* (2007) found that firms' boundary-spanning searching for external resources is greatly helpful to their internal

business activities, especially the boundary-spanning searching in a turbulent environment has positive effects on their innovative activities.

Spatial (boundary-spanning) searching has the following advantages. First, in searching different geographic areas, firms may find novel opportunities such as exploring new market opportunities (Lu and Beamish, 2004; Meyer *et al.*, 2009; Schweisfurth and Raasch, 2018) or reviewing competitors' product strategies. Second, the firms actually failing to engage in geographic extension can acquire knowledge of external production and sales through boundary-spanning geographic search (Ahuja and Katila, 2004; Almeida, 1996). Third, boundary-spanning searching can not only enhance the diversified knowledge collections for knowledge reorganization and innovation (Almeida and Kogut, 1999; Chin *et al.*, 2020) but also link to local knowledge networking (Singh, 2008; Alcácer and Delgado, 2016) and, thus, enhance firms' innovation outputs. In addition, through the above boundary-spanning searching, firms can also acquire complementary or novel local knowledge (Porter, 1990; Nelson, 1993; Cantwell and Janne, 1999) and solve problems (Lahiri, 2010) by effective use of external knowledge (Katila and Ahuja, 2002; Carayannis *et al.*, 2017). All the above mechanisms are benefits brought by boundary-spanning searching. According to this literature, the benefits gained through boundary-spanning searching may lead to new business activities for firms.

According to organizational learning theory (Cohen and Levinthal, 1989; March, 1991) and open innovation perspective (Chesbrough, 2006, 2007; Harison and Koski, 2010), the benefits brought by the above boundary-spanning geographic search can be used by firms to learn business model innovation. The main reasons are: first, a new business model requires firms to integrate internal and external ideas to create value (Holm *et al.*, 2013; Del Giudice and Maggioni, 2014) or through a systematic way with external partners to create or gain value (Osterwalder *et al.*, 2005); second, from the perspective of value creation (Baden-Fuller and Morgan, 2010; Teece, 2010), a new business model must continually update knowledge and expand existing knowledge for organizational routines and processes, to create more value for interested parties of firms (Velu, 2015; Del Giudice *et al.*, 2017). Hence, this study anticipated that, by continually engaging in boundary-spanning geographic search, firms can find new opportunities and learn and obtain the latest ideas and methods for the needs of business activities, which may have positive effects on the creation of valuable new business activities. Based on the previous statement, we propose the following hypotheses:

- H1.* Firms' geographic searching for novel ideas and knowledge acquisition is positively related to business model innovation generations.

Moderating effects of governmental networking

According to the institutional theory (DiMaggio, 1988; North, 1990; Zucker, 1987) and relational governance perspective (Granovetter, 1985; Heide, 1994; Uzzi, 1997), institutions are effective mechanisms to support market operations (North, 1990). However, when institutions fail, informal governance mechanisms (such as social ties) may substitute it and become important in promoting economic and business activities (Peng, 2003). Because social ties offer exchange and coordinate information through informal interpersonal and social mechanisms (Granovetter, 1985), economic and business activities operate smoothly and more efficiently. Particularly, because the market systems are still in the developing stage and face many challenges (Li and Zhang, 2007) such as the emerging China market, require governments to actively participating in and regulating industrial development, guiding industrial policies and affecting firm operations (Hoskisson *et al.*, 2000). Therefore, it is crucial for firms in emerging markets to establish relationships (namely, governmental network) with various government agencies (Li and Zhang, 2007; Hillman *et al.*, 1999). More importantly, firms operating in emerging markets can establish relationships with various government agencies through the governmental network (Peng and Luo, 2000;

Porter, 1990). Because firms have a well-developed governmental network, they can obtain financial aids from the government to search for external novel ideas and knowledge sources. Moreover, governmental networking can help firms to cooperate with local universities, public and private research institutions and science and technology societies, so the governmental network can not only facilitate cooperation with external partners but also create business value by using local science and technology. Finally, governmental networking can assist firms in searching and using cross-regional and cross-national research talents, to create more value for firms. In such a way, governmental networking helps firms to cooperate with local governments and to effectively search and identify valuable knowledge, so as to create more business opportunities.

The local governmental network can increase the legitimacy of local searching and resource acquisition, make firms easier to conduct business activities and reduce illegal imitation and competition (Peng, 2003; Sheng *et al.*, 2011) because governmental networking is a key to getting market information, reducing bureaucratic delay and monetary and non-monetary incentives (Luo *et al.*, 2008). Especially in China, governmental networking supports and dominates business norms and standards (Buttery and Wong, 1999; Park and Luo, 2001). In addition, tight relationships with governments help to strengthen intellectual property protection mechanisms (Kotabe *et al.*, 2014), obtain the necessary protection and appropriate business model innovation. Firms can also address the pressures on economic exchange (resource exchange) networking by strengthening governmental networking (Xin and Pearce, 1996). Governments coordinate and allocate resources in emerging markets to facilitate the business activities of firms (Xu and Zhang, 2008). Hence, a governmental network is important and complement more specific for firms to obtain information and resources from various regions.

In addition, high-quality and diverse market information can be effectively obtained through the governmental network, and firms must understand their customers, and thus effectively adjust their product marketing strategies. On the one hand, governmental networking of firms helps to reduce information asymmetry of emerging markets (Peng and Heath, 1996) because accurate first-hand information can be obtained the fastest through perfect governmental networking. Governmental networking can make firms legitimate and significant to link to relevant government sectors, through which obtain real-time information for business operation and to explore new opportunities or applications on existing products or services. Previous studies also pointed out that a successful business model requires active cooperation between firms and government organizations (Janssen *et al.*, 2008), and firms with tight governmental networking can obtain public resources and enhance the rationality of boundary-spanning geographic search to carry out business activities or assist in new opportunity development.

The joint effects of governmental networking and boundary-spanning geographic search make firms rational in business innovation with the supports from governmental agencies and their institutions, which helps to generate business model innovation. With powerful governmental networking, firms can conduct legitimate boundary-spanning searching to obtain information and intelligence from various regions, so as to make up for information asymmetry in emerging markets and generate exclusive business model innovation:

H2. Firms' governmental networking moderates the relationship between geographic searching and business model generations.

Contingent effects of environment turbulence

While *H2* emphasizes how the governmental network can moderate the relationship between geographic search and business model innovation generations. In *H3*, we further proposed that the joint moderating effects of environmental turbulence and governmental networking in emerging markets affect *H2*. To be more specific, we expected that the

operations in highly turbulent environments would have negative effects on governmental networking and geographic search, as firms' operations and performance are mainly dependent on the fitness of organizations and external environment (Gresov, 1989; Chin *et al.*, 2019), especially in emerging economies with rapid economic growth and rapidly changing industrial environments (Sutcliffe and Zaheer, 1998). Environment turbulence refers to the rapid and uncertain environmental changes faced by organizations (Dess and Beard, 1984). Therefore, in emerging markets, firms must consider the effects of environmental factors in carrying out business activities. Similar to previous studies (Cassiman and Veugelers, 2006; Miller and Friesen, 1983), we assume that firms respond to turbulent environments by acquiring external knowledge resources, especially, the firms running in emerging markets (Zhou and Li, 2010) must consider the effects of environmental factors in their business activities. We proposed that the joint moderating effects of environmental turbulence and governmental networking in emerging markets would weaken geographic search and business model innovation. Previous research has posited that the environment turbulence plays a moderator role in the relationship between organizational factors and outcome variables (Rajagopalan *et al.*, 1993; Moorman and Miner, 1997; Jansen *et al.*, 2005), particularly the effects of external environment on firm innovation (Zahra and Bogner, 1999; Carayannis *et al.*, 2017).

In a turbulent environment, customer behaviors and competitor actions change quickly such as customer preference and competitor imitation and catch-up (Ferrier *et al.*, 1999). Moreover, it is important to detect the valuable opportunities which are unapparent in a highly turbulent environment. Therefore, to discover potential opportunities, besides experience and relevant knowledge, governmental networking is necessary to explore opportunities and acquire resources. In a highly turbulent environment, firms own tight governmental networking not only may obtain first-hand information immediately but also can detect competitors' responses to make an adjustment. A highly turbulent environment helps to seek opportunities but may have negative effects on firms. For instance, D'Aveni (1994) and Santoro *et al.* (2019) pointed out that the firms in highly turbulent environments may bear the risk of operating in outdated ways. Similarly, Jansen *et al.* (2005) found that turbulent environments make it easier for firms' current products and services to be outdated, and they may be easily eliminated by the markets if the latest products and services cannot be developed immediately. Environment turbulence usually reduces the feasibility to acquire innovative resources (Miller and Friesen, 1983; Zahra, 1996), and the pursuit of such high-risk and high-cost innovations may harm firms. Given the emerging market nature of high competition and easy imitation and improvement of existing technologies, such markets may have an increasing impact on firms' business models. We, thus, argued that firms may be impacted by such rapid changes and complex environmental factors of turbulent emerging markets, and may be affected by environment turbulence when acquiring new ideas and knowledge through geographic search:

- H3. High degree of environmental turbulence reduces the joint moderating effects of governmental networking and geographic searching on business model innovation generations.

Method, sample and data

In this study, the questionnaire was designed to obtain the information and to collect firms' data, and the program suggested by Gerbing and Anderson (1988) was used to design the questionnaires. First, in-depth interviews were conducted with six senior managers to understand the industry practice, mainly about how firms carry out a geographic search, governmental networking and business model innovation in current operations. Second, all questionnaires were adjusted and revised based on the previous literature (Sidhu *et al.*, 2007; Spieth and Schneider, 2016) to suitable for our purposes. The questionnaires were first extracted from previous literature, then translated into Chinese by two researchers

proficient in English and Chinese, then translated back into English and finally discussed one by one with ambiguities or conflicts corrected. In doing so, we can make sure all the semantic and sentences achieve consistent meanings. Furthermore, to improve the questionnaires to truly reflect the actual business model innovation, five senior managers were consulted and slight modifications were made according to their suggestions, to ensure that the questionnaires could reflect the condition and background of China, the emerging market, faced by firms. Subsequently, pilot tests were conducted on CEOs of six firms, but the pilot sample sizes were excluded in the final study. The CEOs have completed the questionnaires and provided feedback on item design and language. The questionnaires were refined based on pilot feedback.

The samples of this empirical study were mainly acquired through a large-scale questionnaire survey of the firms in Shenzhen high-technology industrial development zones (HIDs), Zhuhai HIDs and Huizhou Zhongkai HIDs of Pearl River Delta. First, 600 registered firms were randomly selected from the above three high-tech parks. These firms specialized in various manufacturing industries such as electronic parts and computer manufacturing, electronic information and software, bio-pharmaceutical, medical equipment and building materials and equipment. The researchers contacted the CEOs or senior management teams of all firms by phone or email to confirm their willingness to participate in the study and then conducted the questionnaire survey and collection personally. Upon actual contact, a total of 442 firms were willing to participate in this study and a total of 320 firms completed their questionnaires. After incomplete and invalid questionnaires were excluded, a total of 287 firm samples were finally selected for the empirical analysis of this study, with a recovery rate of 47.83%.

Variable measurement

All the variables in our study were based on the previous literature, and the seven-point Likert scale (1 = totally disagree; 7 = totally agree) was used to measure the cognition of CEOs and senior managers of all firms. All the questionnaire scales are exhibited in [Appendix](#).

Business model innovation generations

To examine our primary outcome variable, we anchor on a rich stream of literature that has consistently emphasized the major sources of business model innovation including the customer value proposition, profit formula, key resources and key processes ([Johnson et al., 2008](#); [Moore, 2004](#); [Huang et al., 2013](#)). A business model has an interlocking configuration that taken together, creates and delivers value ([Johnson et al., 2008](#)). In this context, the elements bond to each other in consistent and complementary ways, its power lies in the complex interdependencies of its parts ([Johnson et al., 2008](#)). As argued above, four elements are interdependent and complementary to each other. However, the extant literature is less informative about the weight role of four elements in the business model generation phase. In general, research suggests that four elements are equal to drive business model generation. Thus, we focused on these elements because they provide a configuration mechanism within which four elements have equivalence importance to capture the confluence of business model generations at the firm level. Firms' business model innovation generations were mainly measured by the items developed by [Johnson et al. \(2008\)](#), [Moore \(2004\)](#) and [Huang et al. \(2013\)](#). The business model innovation generations were measured by four items, including profit model, redesign of suggestions on customer value, key resource and process design for new products and services and development of new business development processes without negative effects on core business.

Geographic search

We following the scale proposed by [Sidhu et al. \(2007\)](#) was adopted in this study to mainly measure whether firms obtain information and explore opportunities through boundary-spanning geographic search to make innovations. Geographic searching reflects that firms seek opportunities from various regions through spatial searching. Firms may still obtain external knowledge through boundary-spanning searching even without actual geographic expansion.

Governmental networking

We follow prior literature to capture the firms' governmental networking developed by [Peng and Luo \(2000\)](#), [Xin and Pearce \(1996\)](#). This scale reflects the good relationships between CEOs and senior managers of firms and various government agencies (such as local governments, provincial governments, public and industrial sectors under the jurisdiction of governments).

Environment turbulence

Environment turbulence was measured using five items proposed by [Jansen et al. \(2006\)](#). The environment turbulence was measured to reflect rapid changes, uncertain and highly competitive external markets.

Control variable

We controlled for four firm-level characteristics which may have a different impact on the business model innovation of the firms. The firm size in this study referred to the total number of full-time employees and was measured by the logarithm of the total number of employees of a firm. The firm age was calculated from the establishment to 2020. Research and development (R&D) intensity was used to measure the percentage of annual R&D expenditure in total operating expenses. Various industrial types were controlled in this study because the firms in different industry types have different business model innovations.

Results

Table 1 provides summary statistics and correlations among the variables in our models. As expected, geographic search and governmental networks are positively correlated with business model innovation generation. Moreover, the geographic search is positively

Table 1 Summary statistics and correlations							
Variables	1	2	3	4	5	6	7
Firm age	1.00						
Firm size(ln)	0.56**	1.00					
R&D intensity	−0.16**	−0.18*	1.00				
Geographic search	−0.09	−0.07	0.08	1.00			
Governmental network	0.05	0.10	−0.03	0.28**	1.00		
Environmental turbulence	−0.03	0.05	0.12*	0.37*	0.24*	1.00	
Business model innovation generation	−0.04	0.09	0.06	0.46**	0.26**	0.35	1.00
Mean	16.89	6.24	13.68	5.02	4.18	5.09	4.57
SD	12.89	2.33	16.41	1.35	1.83	1.42	1.50
Cronbach's α				0.85	0.94	0.84	0.96
Composite reliability (CR)				0.91	0.96	0.91	0.97
Average variance extracted (AVE)				0.78	0.90	0.76	0.89
Notes: * $p < 0.1$; ** $p < 0.05$; *** $p < 0.001$							

associated with the governmental network. To estimate discriminant validity for our measures, all constructs were estimated using Cronbach's α , composite reliability (CR) and average variance extracted (AVE); the results appear in [Table 1](#). All the constructs met reliability estimates in our study. Furthermore, the variance inflation factors (VIFs) were calculated for all regressions in our empirical model to test for multicollinearity. Drawing on the [Aiken and West \(1991\)](#) approach, all VIF values were below 2.498, which is lower than the threshold of 10 and indicates that there is no concern regarding multicollinearity issues in our regression model.

To estimate the potential threat of common method variance bias (CMV) to theory, we use Harmon's single-factor procedure to estimate the concern of CMV due to most measurements were self-reported. Based on the approach proposed by [Podsakoff et al. \(2003\)](#), the single-factor test showed that the first factor accounted for 27.16% of the total explained variance. These test procedures suggest that sole source bias is not a concern in our data set.

Hypothesis tests

[Table 2](#) reports the results of the hierarchical regression models used to test our hypotheses. Model 1 is the baseline model that only includes the control variables. Model 2 introduces the control variables and the main effect of the geographic search, which explained a positive and significant coefficient ($\beta = 0.56$, $p < 0.001$) associated with the business model generation. *H1* is supporting. Model 3 introduces the all direct effect of geographic search, governmental network and environmental turbulence on business model generations. As we can find that all the predictor variables are positively and significantly associated with business model generation. Then, we introduce the two-way interaction terms to test our contingency hypotheses. Model 4 introduces two-way interaction effects of geographic search and governmental network on business model generation. As shown in Model 4 of [Table 2](#), the interactive effect of geographic search and governmental network on business model generation is positive and significant ($\beta = 0.06$, $p < 0.05$), offering support for our *H2*. Furthermore, we introduce the three-way interaction of geographic search, governmental network and environmental dynamic to test our hypothesis. Model 5 in [Table 2](#) shows that the three-way interaction of "geographic search $\times$ governmental network $\times$ environmental dynamic" is negative and significant ($\beta = -0.02$, $p < 0.1$) associated with business model generation. The result supports our *H3*.

H3 posits a three-way interaction of geographic search, governmental network and environmental turbulence for business model generation. Following the recommendations of [Dawson and Richter \(2006\)](#), we plotted three-way interaction using one standard deviation below and above the mean. The results for *H3* on the effects of environmental dynamics are presented in [Table 2](#) and [Figure 1](#). Model 5 in [Table 2](#) shows support for *H3* ($\beta = -0.02$, $p < 0.1$). In more dynamic environments, the joint effect of geographic search and governmental network on business model innovation generation is attenuated. The plot in [Figure 1](#) indicates that the use of a governmental network as a protective mechanism across geographic searching is an effective way to catalyze business model innovation generation in both stable and dynamic environments. Yet, a low degree of the governmental network seems to attenuate the effect of geographic search on business model innovation generation, but enhance the effect in a dynamic environment.

Discussion

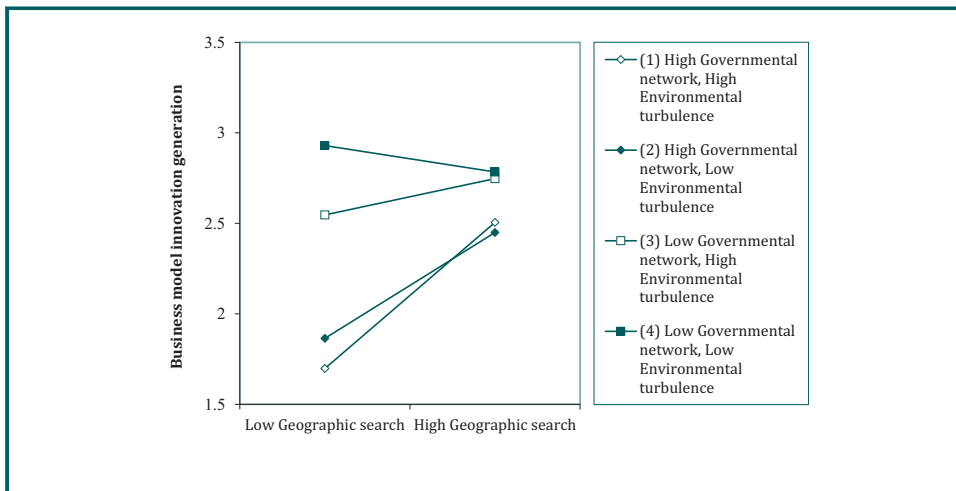
Drawing on the proposed theoretical model, this study investigates the role of geographic search in firms' business model innovation generation on China's emerging economy. We find that geographic search has a stronger positive effect on business model innovation generation. We also sought to gain insight into how governmental networks and turbulent

Table 2 Ordinary least squares regressions models for business model generation

Variables	Model 1	Model 2	Model 3	Model 4	Model 5
Governmental network × environmental turbulence × geographic search (<i>H3</i>)					−0.02* (0.01)
Governmental network × environmental turbulence					0.10* (0.06)
Geographic search × environmental turbulence					0.07* (0.04)
Geographic search × governmental network (<i>H2</i>)				0.06** (0.2)	0.11* (0.06)
Environmental turbulence			0.18** (0.06)	0.17** (0.06)	−0.13 (0.21)
Governmental network			0.09* (0.05)	−0.22 (0.15)	−0.41 (0.29)
Geographic search (<i>H1</i>)		0.56*** (0.06)	0.44*** (0.06)	0.25** (0.11)	0.09 (0.20)
Firm age	−0.01 (0.01)	−0.01 (0.01)	−0.01 (0.01)	−0.01 (0.01)	−0.01 (0.01)
Firm size	0.09* (0.05)	0.10** (0.04)	0.08** (0.04)	0.08** (0.01)	0.08** (0.04)
R&D intensity	0.01 (0.006)	0.00 (0.01)	−0.00 (0.01)	0.01 (0.01)	−0.00 (0.01)
Electronic components and computers manufacturing	−0.40 (0.39)	−0.36 (0.34)	−0.38 (0.34)	−0.22 (0.33)	−0.37 (0.33)
Techcommunication, electrical information and software	−0.19 (0.39)	−0.08 (0.34)	−0.17 (0.34)	0.05 (0.33)	−0.18 (0.34)
Biology and pharmaceutical	0.09 (0.56)	0.14 (0.49)	0.03 (0.48)	0.38 (0.48)	0.07 (0.48)
Financial service professionals	−0.28 (0.45)	−0.35 (0.39)	−0.41 (0.38)	−0.29 (0.38)	−0.36 (0.39)
Construction materials and equipment	0.17 (0.52)	0.34 (0.45)	0.24 (0.44)	0.17 (0.44)	0.25 (0.45)
Medical instruments and equipment	−0.13 (0.64)	0.38 (0.56)	0.27 (0.56)	0.82 (0.55)	0.36 (0.55)
Consultant and service firms	−0.06 (0.45)	−0.06 (0.39)	−0.09 (0.38)	−0.05 (0.38)	−0.05 (0.38)
Constant	4.45*** (0.438)	1.55*** (0.49)	1.01* (0.50)	1.67* (0.76)	2.44** (0.94)
R^2	0.026	0.26	0.29	0.32	0.31
Adj- R^2	0.001	0.23	0.26	0.28	0.27
Model F	0.746	8.81***	8.97***	8.47	7.16***
Observation size	287	287	287	287	287

Notes: * $p < 0.1$; ** $p < 0.05$; *** $p < 0.001$; standardized error coefficients are in parentheses

Figure 1 Three-way interaction of geographic search, governmental network and environmental turbulence



environments operate as contingencies affecting the geographic search-business model innovation relationship. Moreover, our findings reveal that the effects of geographic search are conditional on the governmental network and environmental turbulence. We find that when simultaneously considering governmental network and environmental turbulence in gaining geographic search is decreasing business model innovation generation. This study provides initial evidence that joint moderating effects of governmental network and environmental turbulence is associated with geographic search and business model innovation. This is an important contribution to the business model innovation theory which has not been identified in explaining how and why certainly geographic search exerts their effects on business model innovation generation.

Theoretical implications

Our study advances knowledge in several areas. First, it contributes to the literature about geographic search and business model innovation. Our proposed theoretical framework offers complimentary insights to supply and extend previous studies (Alcácer and Delgado, 2016) that examined geographic search for knowledge resources and business model innovation generation. Specifically, our study offers insight into whether geographic search impact firms' external knowledge access to offer an answer in generating novel business model sourcing. Second, the contribution of the current study to the geographic search literature is the empirical validation of the theoretical argument that a firm's geographic search-business model innovation relationship is moderated by the governmental network. In line with our theoretical arguments, governmental network moderates the geographic search-business model innovation relationship, particularly in a turbulent environment.

Third, as predicted by our theory, the results show that geographic search is an important factor providing a firm with legitimacy and diverse ideas and knowledge for the generation and the implementation of business model innovation. We also show that the governmental network is favorable to a firm's ability to generate an innovative business model, and it presents a contingent approach to the impact of complex geographic search. Importantly, while firms operate in a highly turbulent environment having to cope with a frenetic pace of change under which may detrimental to firms' innovative business model.

In addition to its theoretical significances, we also suggest that managers may use geographic search as a means to stimulate innovative business models, and it describes

contexts where such a search strategy may be most useful in combination with a governmental network. Because the more governmental network that a firm has in weak market institutions (Xin and Pearce, 1996), the more that the legitimacy of geographic searching increases the likelihood that the firm will access external knowledge resources. As such, our study provides a more nuanced perspective on the effects of geographic search on business model innovation generation.

Although we build our hypotheses on the idea that firms' gain the ability to search across geographically as their drive business model innovation, we cannot neglect the well-developed governmental network advantage that state such network brings in emerging market (Hoskisson *et al.*, 2000; Li and Zhang, 2007). As our findings show, a governmental network enables firms to obtain more resources to engage in business model innovation. However, a highly turbulent environment may lead to a dysfunctional mechanism for governmental networks on geographic search. As such, firms are less capable of transforming governmental resources input into business model innovation outcomes. Our empirical evidence sheds new light on how a highly turbulent environment affects the above effects. By considering both the governmental network advantages and environmental turbulence disadvantages associated with the geographic search, our framework provides a more complete understanding of the role of the governmental network in firms' business model innovation activities.

Our findings also offer important implications for managers, who must understand the benefits of governmental networks and the threat of environmental turbulence. A large amount of prior literature has demonstrated that governmental network as an aid to firms' innovation (Kotabe *et al.*, 2014; Wu, 2011; Zhao *et al.*, 2016), but our findings suggest that not only offers an approach for accessing important resources and business model innovation opportunities but also must consider the impact of market environment turbulence. Therefore, firms' managers should develop ties with government agencies and persuade them to offer their crucial resources. By nourishing their governmental ties, firms can increase institutional support from politicians and retain access to crucial resources for innovative business activities.

Limitations and future research directions

This study has a few limitations that suggest directions for future research. First, we conducted our study among three high-technology industrial development zones (HIDs) in the Pearl River Delta Region of China. Future studies are needed to investigate whether our theoretical developing results can be generalized to different sectors and regions in which different types of business model innovation might be salient. Second, we could not analyze the effect of absorptive capability between a firm's geographic search and business model innovation generation. Although we have a contingency effect of the governmental network in our model, which accounts for such geographic search and business model innovation relationships, it would be interesting to examine the effects of absorptive capability. However, future studies could examine alternate absorptive capability measures based on objective data from secondary research such as R&D expenditures, to determine the role of absorptive capability in the governmental network–business model innovation relationship. Third, due to data limitations, we were unable to capture the impact of state-owned on firms' business model generation behavior. Accordingly, an interesting avenue for future research would be to investigate state-owned as control variable according to whether state-owned or privately owned has a significant influence on firms' business model generation activities. Finally, we use survey data and build on a self-reported construct based on previous studies to measure our theoretical proposed, as justified in the measurement section. Although we found that CMV was unlikely to represent a concern to the validity of our results, we acknowledge that the future should possibly collect different sources such as Shenzhen stock-listed companies.

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Appendix

Table A1 Measurement scales^a

Construct and items	SFL
<i>Business model innovation generation</i> Cronbach's $\alpha = 0.960$; CR = 0.97; AVE = 0.893	
Our company has improved customers redesign their value propositions	0.937
Our company has redesigned the company profit formula	0.958
Our company has developed a new business development process without negative effects from core business	0.960
Our company has confirmed key resources and processes to provide products and services	0.924
<i>Geographic search</i> Cronbach's $\alpha = 0.851$; CR = 0.912; AVE = 0.776	
We are knowledgeable about all-important opportunities in the geographic regions in which we operate	0.805
We are well informed about the price and quality aspects of products in neighboring geographic regions	0.926
We closely follow the activities of companies in our industrial sector but operating outside our geographic area	0.907
<i>Governmental networking</i> Cronbach's $\alpha = 0.947$; CR = 0.904; AVE = 0.965	
During the past three years, top managers at our company have heavily used personal ties, networks and connections with: political leaders at various levels of the government	0.958
officials in industrial bureaus	0.960
officials in regulatory and supporting organizations such as tax bureaus, state banks, commercial administration bureaus and the like	0.934
<i>Environment turbulence</i> Cronbach's $\alpha = 0.846$; CR = 0.91; AVE = 0.76	
Environmental changes in our local market are intense	0.875
In a year, nothing has changed in our market	0.886
In our market, the volumes of products and services to be delivered change fast and often	0.862
Notes: SFL = standardized factor loading; ^a Items are measured with seven-point Likert scales (1 = strongly disagree and 7 = strongly agree). CR, construct reliability; AVE, average variance extracted. SFL = standardized factor loading	

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